

PORTFOLIO CONSTRUCTION

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**01 THE BLUEPRINT,
BEFORE THE RULE**

**02 SELECTION, SIZING,
TIMING**

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**04 IN CODE — THEN
STOP**

04

Blueprint before the rule. Your signal becomes a book: the blueprint first, one book in the Lab, the same book in code, then yours.

WHO BROUGHT THEM.

Show the file, not the memory.

01

THE SCREEN,
FINISHED

Keep or drop, counted.

RESULTS.md

02

THE EXAMPLE'S
BLUEPRINT

Read, not run.

init-example

03

YOUR
BLUEPRINT
DRAFT

Drafted, not pushed.

04

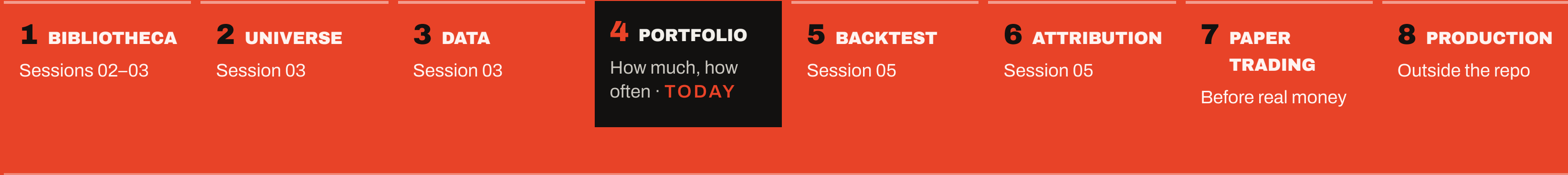
THE DEMO,
WAITING

Requested, not yet
activated.

WHY THE FILE The critic reads your draft today; the push comes before the rule.

SIGNALS IN. WEIGHTS OUT.

Blueprint first. One book in the Lab, the replica in code, then your book.



NOTICE The backtest moved to session 05. Today books are built, not judged.

THE BLUEPRINT, BEFORE THE RULE

the bar, named first · the protocol, written before the test

NAME THE BAR. THEN CLEAR IT.

A JOURNAL_1.md entry, before the blueprint.

01

WHAT “BEAT” MEANS

Index, ETF or equal-weight book of your universe.

02

YOUR MARKET

Same market: the gap measures the rule.

03

PRICED AS

Index or ETF: prices.
Equal weight: weights.

04

EXPERIMENT 1

Plus its control.

REMEMBER Not the benchmark: Experiment 1 is your first rule, measured against one.

WRITE IT BEFORE YOU BUILD IT.

BLUEPRINT_1.md: the hypothesis, fixed once written.

PASTE THIS INTO YOUR ASSISTANT

```
/blueprint 1
```

```
Codex: follow ~/.apm/apm_modules/KaxaNuk/KaxaNuk-Researcher/.apm/prompts/blueprint.prompt.md for  
experiment 1
```

Behind the line: it reads your claims, Bibliotheca/ and RESULTS.md, then writes BLUEPRINT_1.md on your go.

01

WHAT IT PINS DOWN

Thesis, rules, control, predictions, kill switch, statistics, trial budget.

02

EVERY PREDICTION CITES

A note or a measurement. Else, a lead.

03

READ COLD

In Claude Code, blueprint-critic reads it cold, writes nothing.

THEN Push it alone, before the rule.

SELECTION, SIZING, TIMING

Signals identify opportunities. Portfolio construction translates them into capital allocation.

WHO, HOW MUCH, WHEN.

Kept apart, so each can carry a signal of its own.

01

SELECTION

Point in time, delisted included.
Unknown is not no.

02

SIZING

What share each gets. Lag it like
the signal.

03

TIMING

Calendar, event or regime. Every
rebalance is a trade.

THE TRAP Look-ahead. Date t 's book uses $t-1$'s data: one day, at least.

SIX IN THE LAB. FOURTEEN IN THE LIBRARY.

01**NOTHING TO ESTIMATE**

`equal_weight ·
metric_weight(Lab) ·
feature_weighting ·
kn_index`

02**RISK-BASED**

`inverse_volatility ·
risk_parity · hrp ·
herc · max_
diversification`

03**MEAN-VARIANCE, TAIL**

`mean_variance ·
constrained_mean_
variance · black_
litterman · nco · cvar ·
cdar`

04**DECLARE FIRST**

Window, estimator,
linkage: declared, not
tuned.

`blank: 252 · sample · single`

THE BASELINE Equal weight first. Anything else justifies itself on the book.

A CAP FREES WEIGHT. SAY WHO GETS IT.

Your Cash line picks one.

01**REDISTRIBUTE**

Over names with room:
Lab, Weights.cap.

02**OR HOLD AS
CASH**

Trim to a cash row (the
example's: SHY).

03**THE SUM**

1 per date, cash
included.

04**THE DATE**

Dated t, filled at t: the
engine adds no lag.

THE LAB, LIVE

Liquidity-Weighted Trend · six fields · one weights.csv · read the book, not the curve

TREND PICKS. LIQUIDITY WEIGHS.

Claim: sustained uptrend plus deep liquidity beats your benchmark.

01
SELECTION

Signal 1, top 35 by
traded value.

02
SIZING

Cap 20%, excess
spread.

03
TIMING

On change, checked
daily.

04
FEWER THAN 35

Fewer names, fully
invested.

05
NO LAG

Same-day fill: look-
ahead, on purpose.

SIX FIELDS. ONE RULE.

Portfolio → Construction → Metric Weight (cap / value-traded).

01**WEIGHT BY METRIC**

Sizing. Change the default.

`c_daily_traded_value_sma_63d`

02**ELIGIBILITY SIGNAL (0/1)**

Selection.

`c_sma_50d_200d_signal`

03**TOP N BY METRIC**

Selection.

`35`

04**MAXIMUM WEIGHT PER ASSET**

Sizing.

`0.2`

05**REBALANCE FREQUENCY**

Timing.

`daily`

06**REBALANCE TRIGGER**

Timing.

`on_top_n_change`

THE TRAP Frequency left at Select... means none: look-ahead.

READ THE BOOK, NOT THE CURVE.

Open portfolio/<job>/weights.csv: one column
per rebalance.

01**NAMES HELD**

Non-zero cells per column:
regimes.

02**LARGEST WEIGHT**

The top cell, and how many at
20%.

03**HOW OFTEN IT
CHANGES**

Rebalance count, names in plus
out.

IN CODE — THEN STOP

the Lab as answer key · the replica, then your book · same assets, different books · stop before the engine

THE LAB IS THE ANSWER KEY.

Match its arithmetic, not its timing.

01

THE REPLICA

The worked book, in code.

02

THE SIGNAL FIRST

Before any weight.

03

THE WEIGHTS

Within a millionth, cash row dropped.

04

YOUR BOOK

Lagged. Then the five checks.

THE TEST A difference you cannot name is a bug. One you can is a journal entry.

SAME RULE. YOUR CODE.

Save the Lab's `weights.csv` as `Experiment_1/lab_weights.csv`. Then paste.

PASTE THIS INTO YOUR ASSISTANT

- 1 · Using `data-curator-custom-calculations`, add `c_sma_50d`, `c_sma_200d` and `c_sma_50d_200d_signal`, the Lab's names: simple averages of `m_close_dividend_and_split_adjusted`; 1.0 if the 50 is above the 200, else 0.0; null until both exist. Re-run the Curator.
- 2 · Using `portfolio-construction-runs`, in `lab_replica.ipynb`, rebuild `lab_weights.csv` in code as `Portfolio/lab_replica_weights.csv`: no lag, cap spread as the Lab does. Then, in `experiment_1.ipynb`, `BLUEPRINT_1.md`'s rule, as written.

Behind the line: code written fresh in `Data/Curator/` and `Experiments/` — shaped like the example's, never copied.

01

THE COLUMN

`c_*`: one security's own history. Keep the Lab's names: the diff needs them.

SAME ASSETS. DIFFERENT BOOKS.

The KN-TU30 ETF sleeve: an asset-allocation book of six ETFs, with SHY as priced cash.

PASTE THIS INTO YOUR ASSISTANT

```
Using portfolio-construction-runs, size SPY, IEF, TLT, GLD, DBC and UUP, with SHY as cash, on month-  
end rebalances with equal_weight, inverse_volatility, risk_parity, hrp and max_diversification:  
build_allocator per date, long-only, one weight file each.
```

01

WHY ETFS

Sizing is the only decision.

02

DECLARE FIRST

Window, estimator, linkage, calendar. Same for all.

03

READ

Effective positions, by capital and by risk.
Never a return.

ASK Which sizing states the claim? In your strategy, five methods would be five trials.

FIVE CHECKS. EVERY BOOK.

01

**FULLY
INVESTED, AT
MOST**

Above 1: refused.

02

**NO NEGATIVE
WEIGHTS**

Unpinned shorts.

03

**SIGNAL ON THE
PRIOR CLOSE**

Unlagged eligibility.

04

**A FILL PRICE
EXISTS**

Bought before listing.

05

**NOTHING HELD
PAST THE END**

Delisted, still held.

ASK /query which of the six lies does my book risk, and what did we read about it?

STOP HERE, ON PURPOSE.

ON DISK WHEN YOU LEAVE

- `BLUEPRINT_1.md` — pushed alone, before the rule
- `BIBLIOGRAPHY.md` — the blueprint's leads, recorded
- `custom_calculations.py`, `curator.py` — the Lab's signal, a `c_*` column
- `lab_weights.csv`, `lab_replica.ipynb` — the answer key and the replica
- `portfolio_construction.py`, `backtest_engine.py`, `experiment_1.ipynb` — committed
- `Portfolio/portfolio_weights.csv` — your book, lagged, five checks passed
- A `JOURNAL_1.md` entry — every difference named

Left: on disk before you leave. Right: not done, on purpose.

NOT TODAY

- Your alternative sizing — homework, a trial
- Costs and capacity — session 05
- The backtest — session 05. Today you wrote its protocol.
- Attribution — session 05

WHY WE STOP

A replica that agrees, and a book whose every difference has a name, beat a curve you cannot explain.

TO DO.

Commit the code, never Portfolio/.

01

YOUR CONTROL

Same rule, one ingredient out.

Portfolio/

02

TWO LEADS, READ

DeMiguel, Garlappi & Uppal first.

Bibliotheca/

03

YOUR RECONCILIATION

Finish it, if today left it open.

JOURNAL_1.md

04

YOUR ALTERNATIVE

Same names, another sizing.

BEFORE YOU ARRIVE Install the engine and Attribution Analysis. KNBE_API_KEY_KAXANUK, KNAA_API_KEY_KAXANUK in Config/.env. From now on, `uv sync --inexact`.

BACK TO THE PILLARS.

The coding, markets and math sessions behind today. Go back to one when you could not do its part with the AI off.

CODING · CODING FOUNDATIONS

- CF S08 Weight Files and Invariants
weight files, the invariants, Verify
- ▣ CF S06 Time Without Look-Ahead
the prior close: `index < date`
- ▣ CF S09 Design Before You Prompt
the spec before the code
- CF S11 House Rules and Code Review
reviewing the diff against the checklist
- CF S12 Capstone: Ship a Tested Release
`book_check` becomes the Verify cell

MARKETS · FINANCIAL MARKETS

- FM S09 Risk, Diversification and the Long-Only Book
long-only mechanics, two kinds of cash, effective N
- ▣ FM S03 Rates, Bonds and the Yield Curve
SHY as priced cash
- ▣ FM S07 Liquidity, Trading Costs and Capacity
turnover
- FM S06 Indices, Benchmarks, ETFs and Survivorship
TU22-EW as the control book

MATH · MATHEMATICAL FINANCE

- MF S12 Capstone: Overfitting, the Deflated Sharpe and the Holdout
the trial budget and the statistics section
- ▣ MF S05 Linear Algebra and Covariance
covariance and estimation error
- ▣ MF S09 Unsupervised Learning: PCA, Clusters and Regimes
the clustering behind HRP
- ▣ MF S10 Portfolio Optimisation and Estimation Error
`sizing` methods and `sizing.py`

THE READING LIST.

Six leads, none read yet. Read two; start with DeMiguel, Garlappi & Uppal: the case for equal weight as the sizing to beat.

THE PROBLEM · WHY OPTIMISED IS NOT OPTIMAL

MARKOWITZ

1952 Portfolio Selection

MICHAUD

1989 The Markowitz Optimization Enigma: Is “Optimized” Optimal?

DEMIGUEL, GARLAPPI & UPPAL

2009 Optimal Versus Naive Diversification: How Inefficient Is the 1/N Portfolio Strategy?
→ equal_weight

THE ALTERNATIVES · SIZING WITHOUT A RETURN FORECAST

CHOUEIFATY & COIGNARD

2008 Toward Maximum Diversification
→ max_diversification

MAILLARD, RONCALLI & TEÏLETCHE

2010 The Properties of Equally Weighted Risk Contribution Portfolios
→ risk_parity

LÓPEZ DE PRADO

2016 Building Diversified Portfolios that Outperform Out of Sample
→ hrp

ALREADY ON YOUR LIST Grinold & Kahn (2000), ‘Portfolio Construction’; Paleologo (2021), ‘Use Effective Heuristics for Alpha Sizing’.

NOW BUILD IT TWICE.

Once in the Lab, to see it. Once in code, to keep it.

THE LAB

Portfolio → Construction

Answer key: arithmetic, not timing.

THE CODE

lab_replica.ipynb, then
experiment_1.ipynb

Your book, lagged.

BRING

Weight files, blueprint commit hashes.

QUESTIONS research@kaxanuk.mx

NEXT

SESSION 05

TOPIC

BACKTEST & ATTRIBUTION

YOU LEAVE WITH

**ENGINE-PRICED RUNS, A
BRINSON-FACHLER
ATTRIBUTION,
FINDINGS_1.MD.**

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